# Experiment 7
# Planck's Constant

**SAFETY**

Make sure that you have read the **General Safety Notes**, in the Introductory section of this manual, before you begin.

Do not, **under any circumstances**, attempt to repair any of the equipment. If you suspect equipment to be faulty, turn it off at the power point and talk to your demonstrator.

In this exercise, the surface of the globe will become **EXTREMELY HOT**: do **not** touch it, or you will burn yourself. You should handle the globe by using the attached wooden block.

## Outline of Laboratory Exercise

The photoelectric effect is a phenomenon which can occur when light hits a metallic surface. If the photons have enough energy they can eject electrons from the metallic surface and cause a current. This phenomenon was first described by Albert Einstein for which he received the Nobel Prize. This laid the important foundation for the quantized nature of light.

You can read about the photoelectric effect here:
http://hyperphysics.phy-astr.gsu.edu/hbase/mod1.html

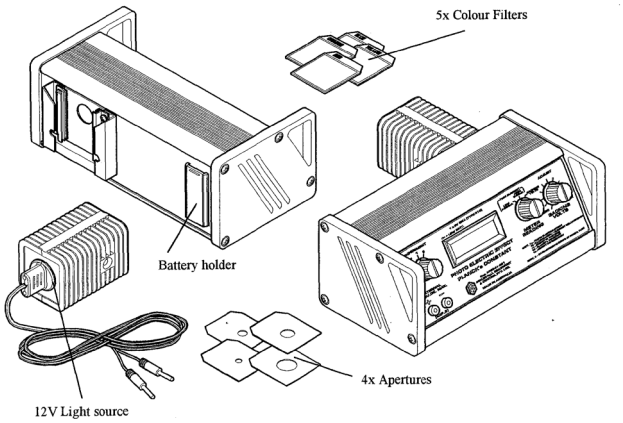
✓ ***Pre-Lab Exercise***: *Read the entire laboratory exercise. You must pay particular attention to the relevant Safety Notes for this exercise, as well as reviewing all the theoretical equations given below. Complete the online pre-lab exercises.*

## Learning Goals

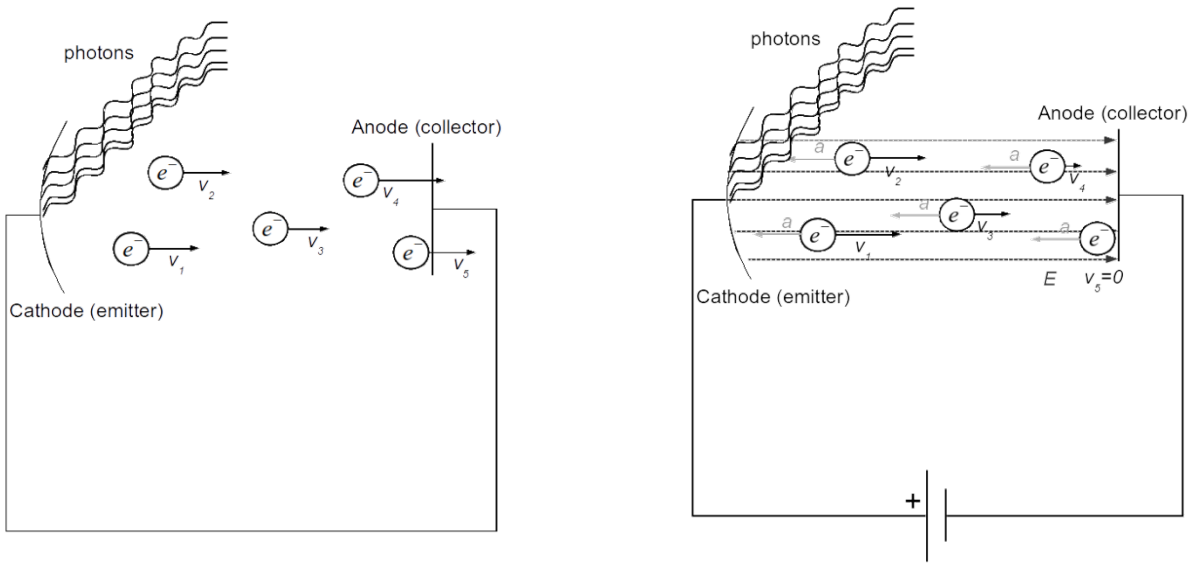
- To carefully collect data and determine a relationship to define certain physical parameters and relate to the experiment and physical phenomena occurring.
- To proficiently use EXCEL when creating and analysing data.
- To formulate an understanding of the mechanisms when light is incident on matter or matter creates light.

## Equipment

The equipment on your desk should be a black box which looks something like this:




Inside the black box is a simple circuit with a battery and a **photocell** – a device which turns light into electricity. The simplified circuit diagram is shown below with and without a voltage applied.




**Referring to the left hand diagram,** when a single photon strikes the metal surface of the cathode (the curved metallic plate), the energy of the photon is absorbed by a single electron on the metal surface. This will give energy to the electron and allow it to move between energy levels in the electron shells. If the photon has enough energy the electron will overcome the potential of the nucleus and be emitted from the atom.

The minimum energy that the most loosely-bound electron must absorb from a single photon in order to break free from the metal is called the metal's **work function, W**. The value of this work function will change according to the physical qualities of the metal. While light is falling on the cathode, billions of electrons per second will be emitted and some of these electrons will hit the anode. This creates a small flow of electrical current.

**Referring to the right hand diagram**, if we apply an external voltage to the circuit, in the **opposite** direction to the flow of charge, then the voltage will work against this flow. If this 'backing voltage' is powerful enough, the flow of electrons will stop completely: the energy that 'pushes' the electrons to flow in one direction will be cancelled out by the voltage that 'pushes' in the opposite direction. The minimum voltage required to stop all the electrons from flowing – i.e. to reduce the flow of current to zero – is called the **stopping voltage $V_s$**. With the voltage the electron with the largest kinetic energy, $KE_{max}$, will have stopped just before the anode and start to accelerate back towards the cathode.

## Energies and Voltages

According to Einstein's 1905 model, light is made up of particle-like 'packets' or 'quanta' of energy called **photons**. The energy of a single photon is proportional to the frequency $f$ of the light:

$$E = hf,$$

where $h$ is Planck's constant.

Electrons emitted from the metal move at different speeds. If the *fastest* electrons (those with the greatest kinetic energy) are barely stopped by the applied backing voltage, those electrons must lose an amount of kinetic energy exactly equal to their gain in electrical potential energy as they move through a potential difference equal to the stopping voltage. That is

$$KE_{max} = \frac{1}{2}mv^2 = eV_{stop}$$

where $e = 1.602 \times 10^{-19}$ C is the magnitude of the electron charge. Recall that the change in electrical potential energy for a charge $q$ moving through a potential difference $V$ is $\Delta U = qV$.

When a photon is absorbed by an electron in the metal, the electron absorbs all of the photon's energy. As the electron escapes from the metal, it uses some of the absorbed energy to break free. The *minimum* energy required to escape is the work function, $W$. Any energy left over is kinetic energy of the electron. Therefore

$$hf = W + KE_{max}$$

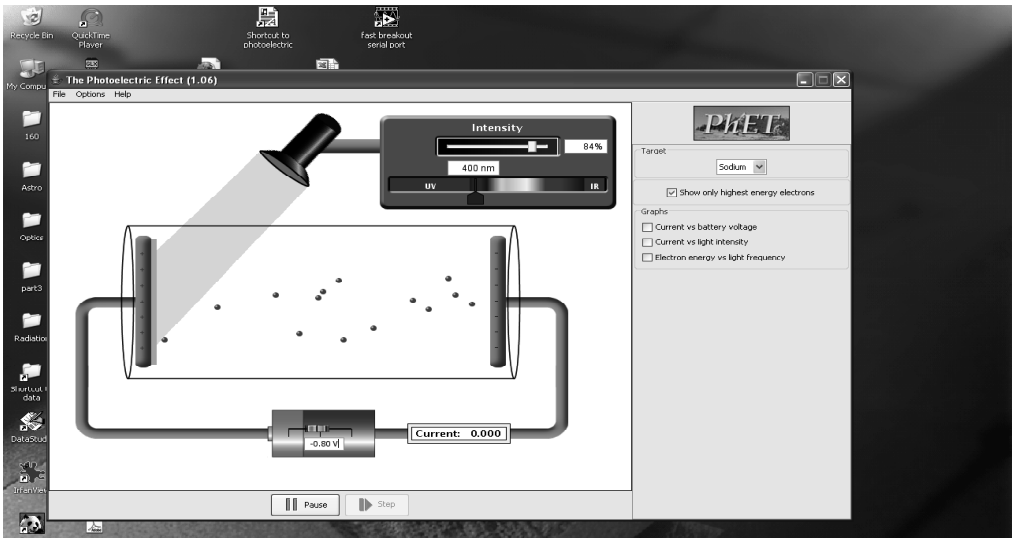
Since $KE_{max} = eV_{stop}$ we have

$$V_{stop} = \frac{hf}{e} - \frac{W}{e}.$$

This means that if we know the frequency of the light as well as the stopping voltage, we can determine the work function of the metal being used as well as Planck's constant.

## Verifying the Photoelectric Effect

First, open the photoelectric simulation file (it should be on your computer's desktop, if not it can be found here: https://phet.colorado.edu/en/simulation/photoelectric).



By playing with this software, you should develop a feel for what is happening. In the program, you should:

- Select **Sodium** as the target
- Turn up the intensity (of the lamp)
- Select only the highest-energy electrons
- Adjust the voltage until the electrons just stop before hitting the anode

You should require a **negative** voltage to achieve this.

✓ What happened during the simulation?

Adjust the settings so that current is flowing. Now change the target from **Sodium** to **Platinum**.

✓ What happens to the simulation?

With Platinum as the new target, steadily increase the frequency (i.e. decrease the wavelength) of the light.

✓ Describe what happens. What does this tell you about the difference between Sodium and Platinum, and their respective work functions?

When you are finished playing with the software simulation, it is time to use the photoelectric equipment on your desk. Make sure that the light source (the lamp) is fitted securely, and turn the equipment on.

**SAFETY NOTE**

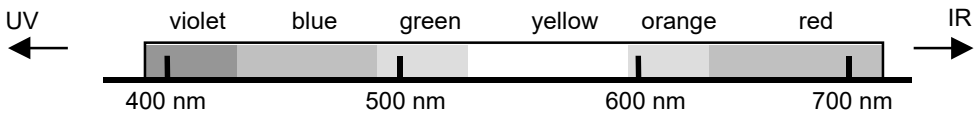
The light source will become very hot. This is normal. However, do **not** leave the light on and unattended for too long, or overheating may occur. Be careful!

Set up your equipment as follows:

- Select 'Experiment 1'
- Set the meter switch to read BACKING VOLTAGE
- Turn the ADJUST knob anti-clockwise to set the Backing Voltage to zero
- Set the meter switch to read TUBE CURRENT in microamps. The **tube current** is a measure of how many electrons are passing from the cathode (metallic plate) to the anode (collector) per second – i.e. how many electrons are absorbing photons and undergoing the Photoelectric Effect. Because the light source is currently off, this should display approximately zero.
- Insert the aperture disk with the largest aperture.
- Turn on the 12 V light source.

As well as several different aperture disks, the equipment also has five coloured filters. These filters are designed to block some colours (i.e. frequencies) of light but to let other colours through. For instance, the blue filter will only let through light that has a wavelength of 428 nm.

An approximate relationship between colour and wavelength is as follows:



By applying a negative voltage to your circuit you should be able to reduce the tube current to zero, just as in the software simulation. Follow the directions below (record your data using Excel):

- Select one of the coloured filters and place it in the apparatus.
- Slowly increase the backing voltage. The tube current should slowly decrease to zero.
- When the tube current reaches zero on the most sensitive range (you will need to switch down from µA to nA range), turn the meter switch to BACKING VOLTAGE and record this value of the backing voltage. This value should be the **stopping voltage**.
- Repeat this 3 times, and find the average value for the stopping voltage.
- Repeat the directions above to find the average stopping voltage for each different coloured filter.

Using Excell, plot your data. Consider what your axes should be in order to obtain an experimental value of Planck's Constant h.

**Analysis**

- ✓ Comment on your plot and analyse your data.
- ✓ From this graphical analysis of your data, find the work function.
- ✓ From the graphical analysis of your data, find Planck's Constant **h.**
- ✓ Compare the experimental value of h to the actual value $6.626 \times 10^{-34}$ J s (or equivalent, $4.136 \times 10^{-15}$ eV s, recall too that 1eV = e*1V).